Al-Naser Al-Rais

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EDUCATION

University of Cambridge **Cambridge, UK**
BA in Natural Sciences *Oct 2024 - June 2027*

- **Courses:** Biochemistry, Computational Biology, Pharmacology
- **Coursework**: Cell and molecular biology, protein biochemistry, drug design and pharmacodynamics, mathematical biology
- **Accomplishments: 1 of 10 worldwide** selected for fully funded placement at TU Munich (Max Planck Institute); Vivensa Foundation Ignition Fund Fellow (**£5,000 grant** for aging research); Raised **£6,000** to fund aging research in Church lab; G.R.N. Minchin Award (substantial travel grant funded by Cambridge); Bio-Spark Exploring Fellow

EXPERIENCE

Knowles Lab, University of Cambridge **Cambridge, UK**
Undergraduate Researcher, Chemistry *Oct 2025 – Present*

- Fabricated PDMS microfluidic devices using soft lithography for peptide hydrogel screening
- Built UV spectroscopy calibration curves and measured peptide hydrogel dissociation kinetics across pH/salt conditions
- Characterising full length-Tau monomer states (coil-like vs premolten globule-like) using circular dichroism spectroscopy
- Building a microelectrophoresis platform for single-particle charge determination of LNPs and condensates

Lu Lab, Harvard Medical School **Boston, MA**
Undergraduate Research Fellow, Systems Biology *Jun 2025 – Sep 2025*

- Engineered doxycycline-inducible protein aggregation model in HEK293T and U2OS cells
- Expanded model to include Tau, alpha-synuclein, TDP-43, FUS via Gibson assembly
- Generated stable cell lines expressing sfGFP-tagged constructs using lentiviral transduction
- Created 20 P301X Tau (K18 domain) variants using site-directed mutagenesis to map aggregation propensity and toxicity
- Quantified aggregate formation and cytotoxicity using fluorescence microscopy and cell detachment counts

EXTRACURRICULAR ACTIVITIES

No Patient Left Behind
[Fellow] *Mar 2026 – present*

Cambridge Society for Ageing and Longevity Research
[Head of undergraduate engagement] *Jan 2026 – present*

- Organised 100-person screening of *Forever Young*, hosting Steve Horvath

Darwin Society (Christ's College Scientific Society)
[Speaker's Officer] *Jun 2025 - present*

- Organised high-profile speaker events and formal dinners

British Powerlifting
[National Competitor, 2022]